

SMALL BUSINESS INNOVATION RESEARCH (SBIR / STTR) PHASE I PROPOSAL

PROJECT: Deterministic SMT AI Guardrails & High-Throughput Microgrid Command Center

1. Executive Summary & Technical Merit

Emerging Defense Solutions (EDS) proposes the deployment of a defense-grade, high-performance SOC Command Center AI engine. By integrating real-time formal verification via Z3 SMT solvers into local LLM token generation streams, the proposed technology guarantees zero Controlled Unclassified Information (CUI) spillage ($\$P_{\text{violation}} = 0\$$) inside AMD SEV-SNP Guest TEE enclaves while maintaining speculative high-throughput performance.

2. SBIR / STTR Budget & Phase I Allocation

Phase I Category	Allocation (\$)	Technical Focus Area
R&D Engineering (SBC)	\$165,000	Monad Logits & PyTorch SDPA Kernel Optimization
Research Institution (STTR)	\$85,000	Formal Z3 Logic Solver & Verification Proofs
Hardware Enclave Testing	\$35,000	AMD SEV-SNP Guest TEE & Cerebras CS-4 Emulation
Direct Overhead & Fringe	\$20,000	Compliance, Security Boundary Audit & Documentation
TOTAL PHASE I REQUEST	\$305,000	NSF / DoD Phase I Grant Limit Target

3. Commercial Potential & Dual-Use Impact

Defense Applications: Real-time automated SOC intelligence for C4ISR tactical enclaves, air-gapped command posts, and zero-trust data boundaries.

Commercial Applications: Datacenter microgrid optimization, heat exhaust recovery for sustainable agriculture, and enterprise LLM inference with zero data leakage.